

Ubuntu Linux Administration: User and Group Management

Understanding User Accounts in Linux



Andrew Mallett

Author and Trainer

@theurbanpenguin www.theurbanpenguin.com



Overview



Managing Users and Groups in Linux

Course Overview

- Privilege escalation
- Creating user accounts
- Managing passwords in Linux
- Managing system-wide login scripts
- Using PAM

In this Module

- Understanding Linux users
- Using sudo
- Using su
- Using pkexec





Lab Setup

Ubuntu Linux Administration: Essential Commands

Andrew Mallett





Linux Users

Database

- `/etc/passwd`
- `getent passwd`
- `getent passwd <username>`
- `man 5 passwd`
- `/etc/nsswitch.conf`

Types

- standard users
- system users
- UID 0 (root)

Case-sensitive username as well as password



Demo



Investigate Linux Users:

- User account database
- Name service switching
- Listing users and filtering using *AWK*





UID 0, usually root, manages your Linux System.
Using **su**, **sudo**, **pkexec** we can elevate to UID 0

```
$ su
```

```
$ su -
```

```
$ su -l
```

```
$ su - bob
```

su – Substitute User

Using the command **su** we can change to the root account, or any account that we know the password. If the root account does not have a password, as is the case with Ubuntu, then **this is not an option unless we set the root password.**

Having all administration team
all knowing the root password
is not the best of ideas!



Demo



Let's begin by using the su command:

- Login or non-login shell
- Testing user accounts



```
$ sudo less /etc/sudoers  
$ sudo ls /etc/sudoers.d/
```

Sudo

Using sudo, we can delegate administrative tasks without the need to divulge the root password or give access to all commands

```
$ sudo visudo
```

```
$ sudo visudo -f /etc/sudoers.d/bob
```

Editing the Sudoers Files

Using the command `visudo` to make changes to the configuration will enforce a syntax check when the file is saved.

A misconfigured sudo entry will disable sudo access to your system.

```
bob ubuntu1=(root) NOPASSWD: ALL
%sudo ALL=(root) ALL
%helpdesk ALL=(root) /usr/bin/passwd, !/usr/bin/passwd root
```

Sample Entries

We illustrate 3 entries that you may be able to use

Demo



Understanding Sudo:

- `/etc/sudoers`
- `/etc/sudoers.d/`
- adding a safety-net with admin group
- editing sudoers file



```
$ sudo cat /etc/polkit-1/localauthority.conf.d/51-ubuntu-admin.conf
$ echo $$

$ pktttyagent --process 5296

$ pkexec cat /etc/shadow

$ enter password
```

Using Polkit (Formerly PolicyKit)

Designed more for desktop systems we have PolKit. A separate authenticator program is required to authenticate users. Make sure your user has a password and you belong to the wheel group. We require the authenticator to monitor the process in another window. Make sure your user is in the **sudo or **admin** group.**

Demo



Using a default PolKit rules

- use pktttyagent and pkexec
- recover from failed sudo system



Summary



In this module we have introduced Linux users and ways to elevate privileges

- getent and /etc/nsswitch.conf
- awk
- su
- su -
- su -l
- sudo
- visudo
- visudo -f
- pkexec / pktttyagent



Creating and Managing Users in Linux

